

DATA COLLECTION & DATA SOURCE IDENTIFICATION

Data Analytics Internship – Task 1 Report

1. Introduction

Data collection is the first and most important step in any data analytics process. The quality of analysis depends on the reliability and accuracy of the data used. This task focuses on identifying a suitable dataset and understanding its source and structure.

2. Objective

The main objectives of this task are:

- To identify a relevant dataset for analysis
- To understand different types of data sources
- To perform initial data inspection
- To verify data quality and structure

3. Dataset Selection

3.1 Dataset Name

Student Performance Dataset

3.2 Source

The dataset is obtained from the Kaggle platform, which provides publicly available datasets for learning and research.

3.3 Type of Data

The dataset is classified as **secondary data**, as it has already been collected and published by another source.

3.4 Data Format

The dataset is available in **CSV (Comma Separated Values)** format, which is widely used for data analysis.

4. Dataset Description

The dataset contains structured information related to student performance. The key attributes included are:

- Student_ID
- Gender
- Study_Hours
- Attendance
- Marks

This dataset can be used to analyze the relationship between study habits and academic performance.

5. Methodology

5.1 Data Collection

The dataset was downloaded from a public repository and stored locally in CSV format.

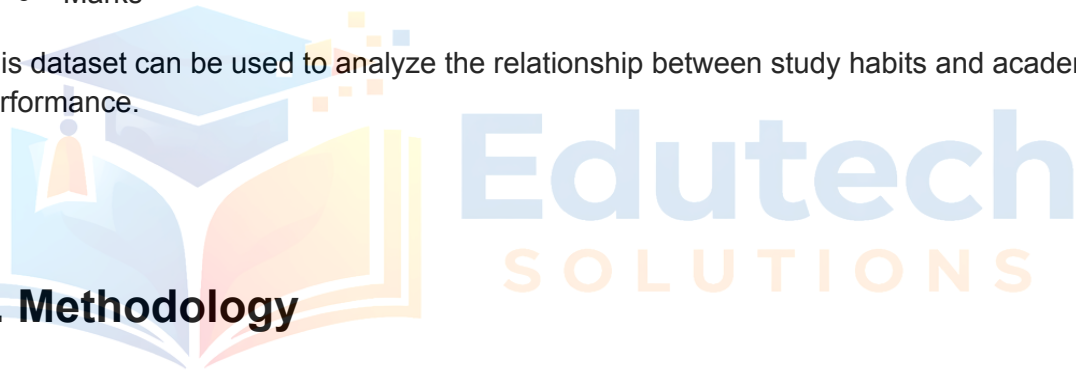
5.2 Data Loading

The dataset was loaded into Python using the Pandas library.

5.3 Data Inspection

Basic data exploration techniques were applied, including:

- Viewing initial records
- Checking dataset structure
- Generating summary statistics
- Identifying missing values



6. Implementation

The following Python code was used for data inspection:

```
import pandas as pd

df = pd.read_csv("student_data.csv")

print(df.head())
print(df.info())
print(df.describe())
print(df.isnull().sum())
```

7. Results and Observations

- The dataset contains structured and clean data
- No missing values were found
- The dataset includes both numerical and categorical variables
- Basic statistical analysis provides insights into student performance

8. Key Concepts

8.1 Primary Data

Primary data refers to data collected directly from original sources such as surveys, experiments, or observations.

8.2 Secondary Data

Secondary data refers to data that has already been collected and is available for use, such as datasets from online platforms.

9. Common Data Formats

The most commonly used data formats in data analytics include:

- CSV
- Excel (XLSX)
- SQL Databases

10. Conclusion

This task successfully demonstrated the process of identifying and collecting a dataset from a reliable source. Initial data inspection confirmed that the dataset is clean, structured, and suitable for further analysis. Understanding data sources and formats is essential for effective data analytics.

11. Future Work

- Data visualization using charts and graphs
- Advanced data analysis techniques
- Predictive modeling using machine learning

